NCATS OLIGOTOX OPEN DATA CHALLENGE · PHASE 2

OligoTox-Kidney

A curated, per-measurement **kidney-toxicity** dataset for oligonucleotide therapeutics

65 oligos · 111 measurements · 35 target genes · 100% kidney-specific

Prepared for scientific review — German (biochemist) · June 2026

THE IDEA IN ONE SENTENCE

Oligonucleotide drugs can injure the kidney in a way the standard safety test misses — so we built a dataset that captures the right signal, one experiment per row, every value traceable to its source.

Why this matters

Oligonucleotides are a booming drug class — ~19 already approved, dozens in trials, treating diseases nothing else could.

But the kidney is a common casualty. The kidney filters and concentrates these drugs, so it is exposed to high levels — and several oligos have caused proteinuria, kidney injury, even kidney failure.

The catch: the damage is often *invisible* to routine cell-survival screening (next slides).

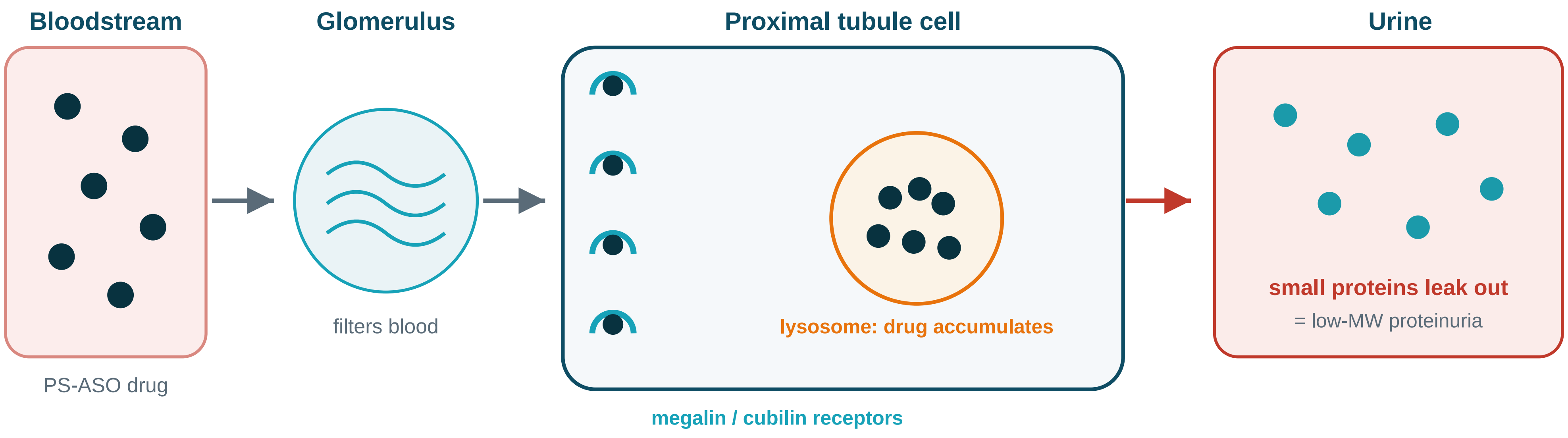
The gap we fill
There is no open, well-structured dataset that pairs each oligo's **design** (sequence + chemistry) with a **graded kidney outcome**. That is exactly what a model would need to predict risk early — and what we assembled.

PART 1

The science: a quiet kind of kidney injury

Why a cell-survival test gives these drugs a false "all-clear"

How these drugs injure the kidney



Reabsorption FAILS

Protein leaks to urine — but reversible.
Functional injury → **GRADE 1**

Cell SURVIVES

Viability normal, no cell death.
A viability test sees **GRADE 0**

Phosphorothioate ASOs are filtered, then vacuumed into proximal-tubule cells by the megalin/cubilin receptors and stored in lysosomes. Reabsorption of small proteins fails → they leak into urine (proteinuria) — yet the cell stays alive.

The trap: one drug, two tests, opposite answers

Same drug. Two tests. Opposite conclusions.

Viability / MTT test

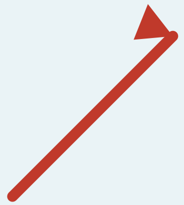
"Are the cells still alive?"

Cells alive ✓

VERDICT: "looks clean" (misleading)

Functional test

"Is the cell still doing its job?"



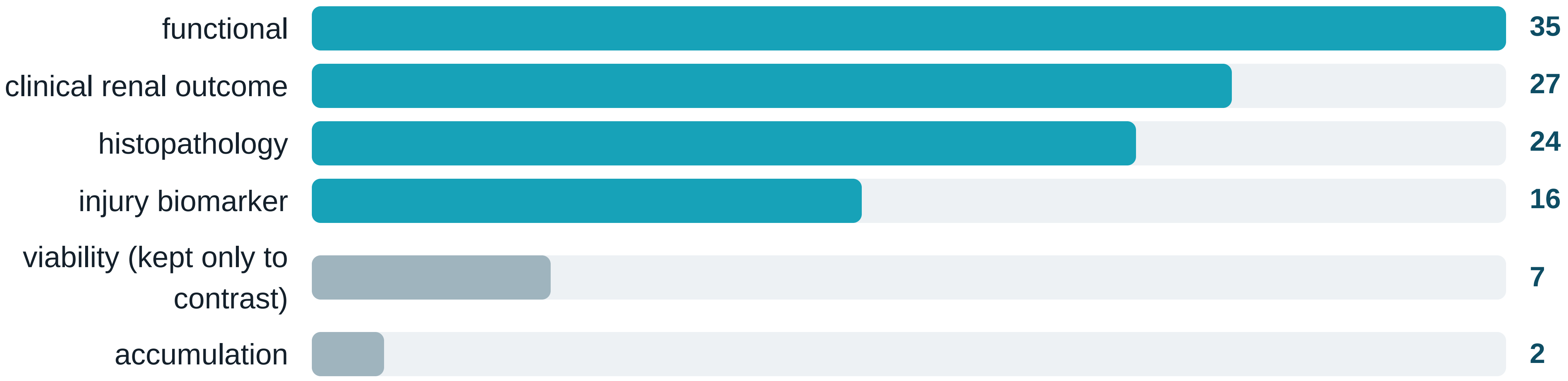
Protein reabsorption fails → proteinuria

VERDICT: real toxicity (GRADE 1)

A viability/MTT screen asks only **"are the cells alive?"** — and these drugs don't kill cells. So routine screening calls them **clean**. Our schema is built to capture the **functional** signal that test misses.

So we record the *right* readouts

We deliberately weight the data toward **function and injury markers**, not cell survival:



“Readout” = one measured thing. Biomarkers = chemicals (KIM-1, NGAL, clusterin, cystatin C) that rise when the kidney is stressed. Functional + biomarker rows (51) dwarf viability rows (7) — by design.

PART 2

What we built

The dataset, its structure, and how toxicity is scored

The dataset at a glance

65
unique oligos
(7 drug families)

111
graded measurements
(target was ≥100)

100%
kidney-specific
(no padding)

16
distinct sources
fully cited

Spans every oligo modality, all three study types (dish / animal / human), and the full severity range from safe to severe — including deliberate negative controls.

35 target genes · **33/65** sequences filled (rest honestly marked "TBD", never guessed) · every row carries its source.

How the data is organised — two linked tables

65 drugs

data/oligos.csv

one row per drug · the PREDICTORS (causes)

🔑 oligo_id (primary key)

oligo_class · target_gene

backbone_chemistry · sugar_mods

gapmer_design · conjugate

ps_count · length_nt

sequence_5to3 · max_phase

17 columns describing the drug's
sequence + chemistry + design

oligo_id

links them

111 measurements

data/measurements.csv

one row per result · the OUTCOMES (effects)

🔑 measurement_id (PK)

🔗 oligo_id (foreign key)

study_type · species · system_model

delivery · dose · readout_name/value

nephrotox_grade (0–3)

source_id · source_ref · redistribution

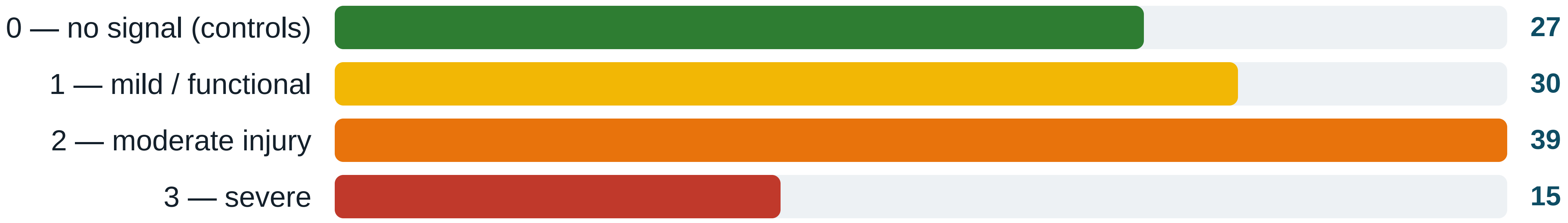
23 columns: the graded result +
a full source trail for every row

One table describes each **drug** (the suspected causes); the other lists each **result** (the effects). They share an ID (`oligo_id`) so any result links back to its drug — like two tabs in a spreadsheet joined by a key.

The scoring system: a 0–3 severity grade

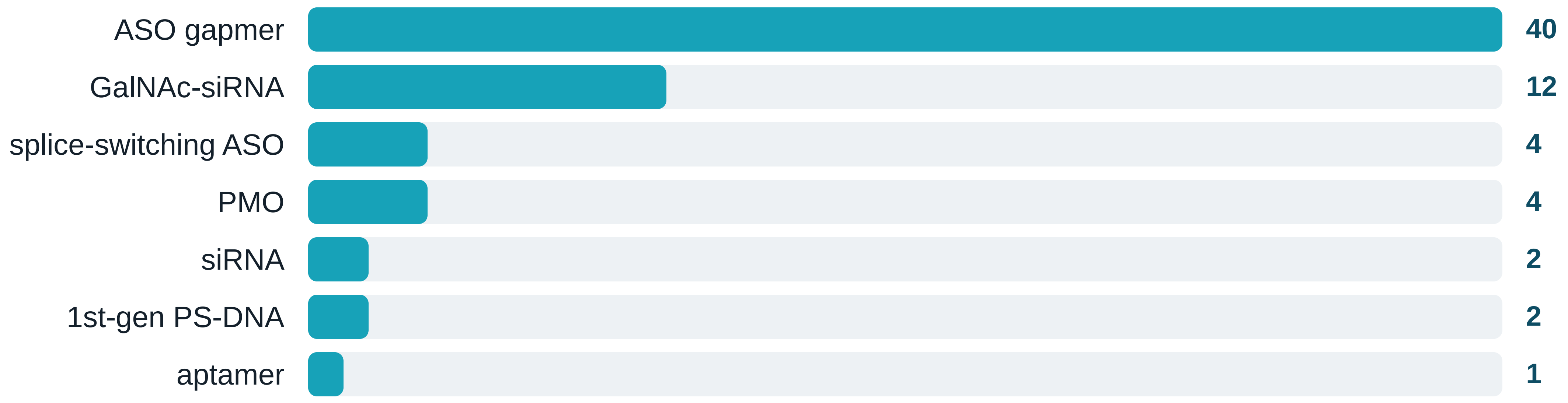
Each result gets one ordinal grade by a written rubric (in `schema.md`). **All grades are currently flagged “provisional”** — confirming them is the sign-off we’re asking German for.

Grades span the full range — including safe controls



A healthy spread across all four levels — a usable dataset needs examples of **every** severity, not just the dramatic ones. The 27 grade-0 rows are real negative controls (GalNAc-siRNA, intrathecal ASO, aptamer).

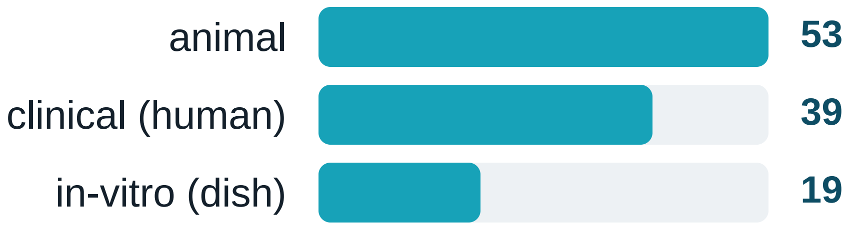
Coverage: every oligonucleotide family



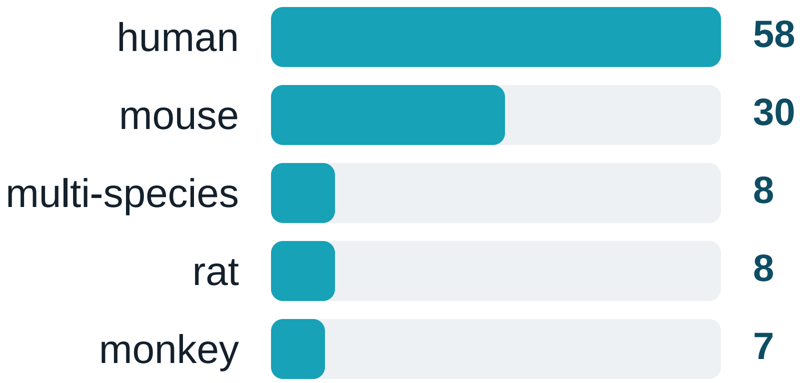
Gapmer = central DNA “gap” + chemically-modified wings (cuts target RNA). **siRNA** = double-stranded silencer; **GalNAc** = sugar tag aiming it at the liver. **PMO** = neutral-backbone splice fixer (the muscular-dystrophy drugs). **Aptamer** = an oligo folded to grab a protein, like an antibody.

Coverage: dish → animal → human (the translation axis)

STUDY TYPE



SPECIES



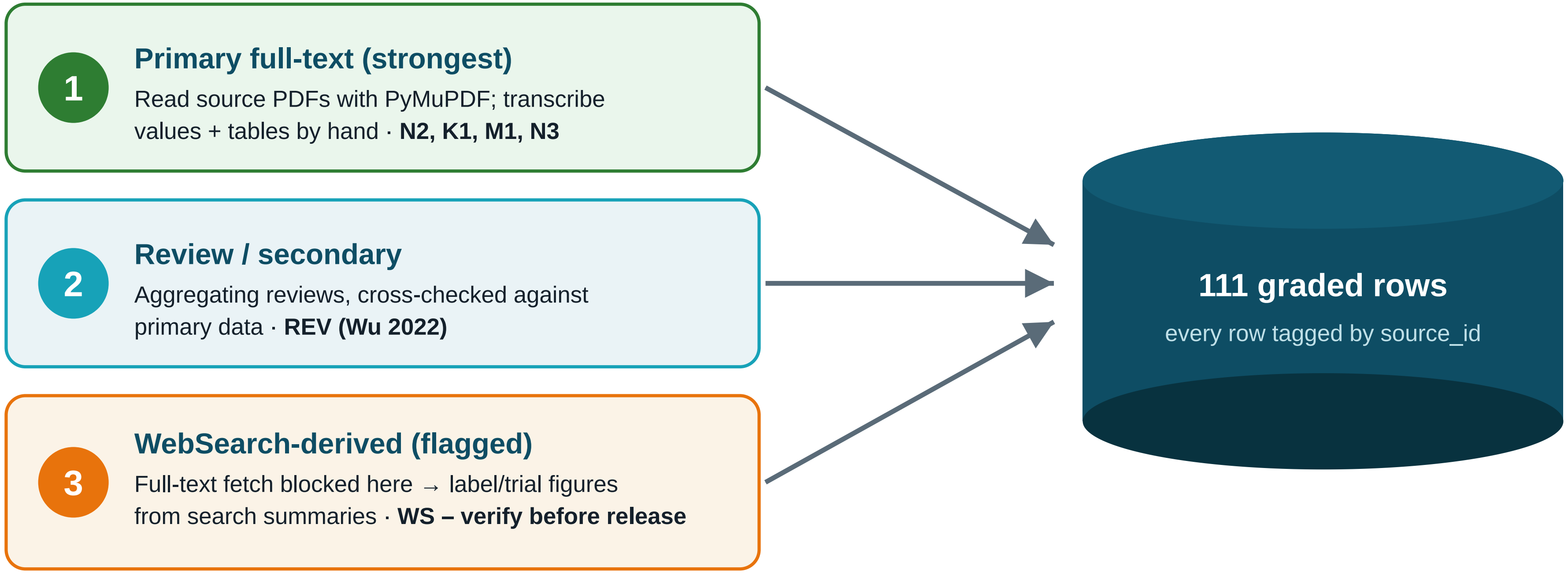
Having dish, animal, and human rows side by side — measured the same way — lets the **animal-to-human translation gap** be studied directly (Finding 3).

PART 3

How we built it

Methodology, the integrity rule, and the sources researched

Three ways we gathered data — each tagged per row



Every row records *how* it was obtained, so its reliability is visible. Web-search rows are flagged for re-checking because this secure environment blocks downloading full papers.

The rule that makes the dataset trustworthy

⚠️ Strict no-fabrication policy

Drug **sequences** and toxicity **values** are never invented or recalled from memory. A value goes in **only** if an explicit, citable, redistribution-permitted source states it. Drugs with no published kidney data were **omitted, not padded**.

WHY SEQUENCES ARE 33/65, NOT 65/65

The 32 blanks are **real gaps**, honestly marked “TBD”. We could have made the table look complete by guessing — but a falsely-complete table can’t be trusted at all.

WHY THIS MATTERS TO A REVIEWER

If the easy fields were fudged, none of the hard toxicity values could be believed. Honest blanks are the **credibility** of the whole dataset.

Sources researched

STRICT-KIDNEY PRIMARY (READ IN FULL)

- **Janssen 2019** · drisapersen, human kidney-cell injury *(N2 · 10 rows)*
- **US 11,105,794 B2** · patent tox panel *(N3 · 21 rows)*
- **Moisan 2017** · human tubule-cell panel *(M1 · 11 rows)*
- **Sandelius 2020** · urine injury-biomarkers *(K1 · 9 rows)*
- **van Poelgeest 2013** · SPC5001 first-in-human AKI *(A3)*
- **Arch Toxicol 2021** · SPC5001 kidney-on-chip *(A4 · 5 rows)*

ANCHORS · REVIEW · PATENTS

- **Wu 2022** · marketed-ASO nephrotoxicity review *(REV)*
- **inotersen** · NEJM NEURO-TTR + FDA label — severe anchor *(A1)*
- **mipomersen, volanesorsen** · gapmer renal signals *(A9, A8)*
- **inclisiran, givosiran, nusinersen** · safe controls
- **US 11,479,818 B2** · 2nd patent, queued *(N4)*

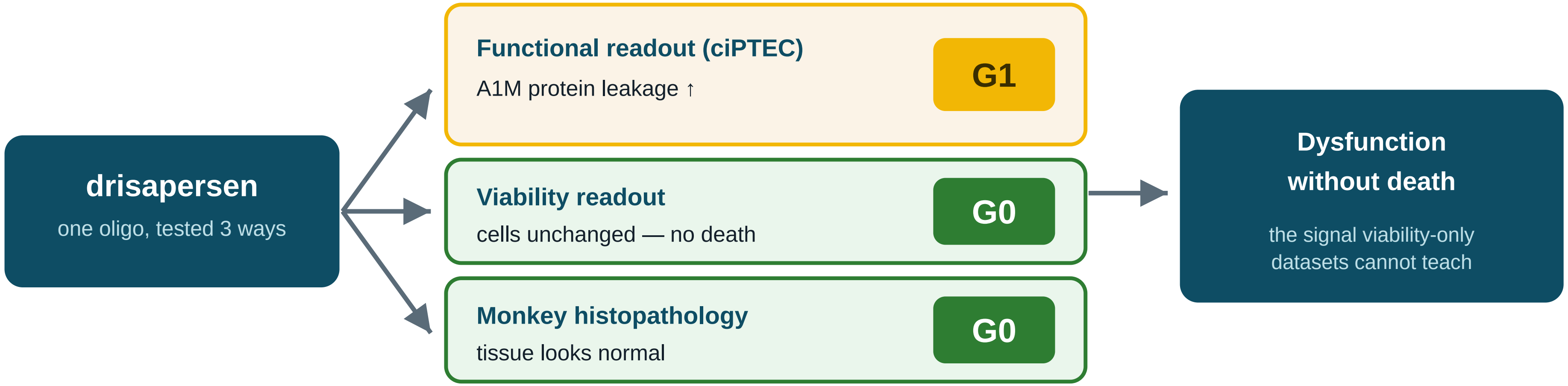
+ 36 “WS” rows from FDA/EMA labels & pivotal trials: patisiran, vutrisiran, lumasiran, nedosiran, eplontersen, tofersen, bepirovirsen, olpasiran, the DMD PMOs, pegaptanib, fitusiran, zilebesiran — plus Crooke 2018 (human) & Yu 2012 (monkey) for the translation gap.

PART 4

What the data shows

Three findings worth a biochemist’s attention

Finding 1 — the “invisible” injury, captured



For the **same** drug we store **paired rows**: a mild grade on the functional test, a clean grade on viability/tissue. Side by side they encode “dysfunction without death” — a distinction viability-only datasets simply cannot teach a model.

Finding 2 — a patent unlocked sequence + toxicity together

US 11,105,794 B2 — Table 1 (public domain)

compound	sequence	SEQ ID	in-vivo class
Cmpd 1-1	CGTcag…AATC	1	low → G1
Cmpd 1-2	ACGtta…GGTA	2	medium → G2
Cmpd 2-1	TGCcat…TACG	3	high → G3
…	…	…	

sequence AND toxicity in the SAME row — rare & valuable
words mapped to our 0–3 scale

+21 measurement rows

our single biggest source
in one clean, public-domain panel

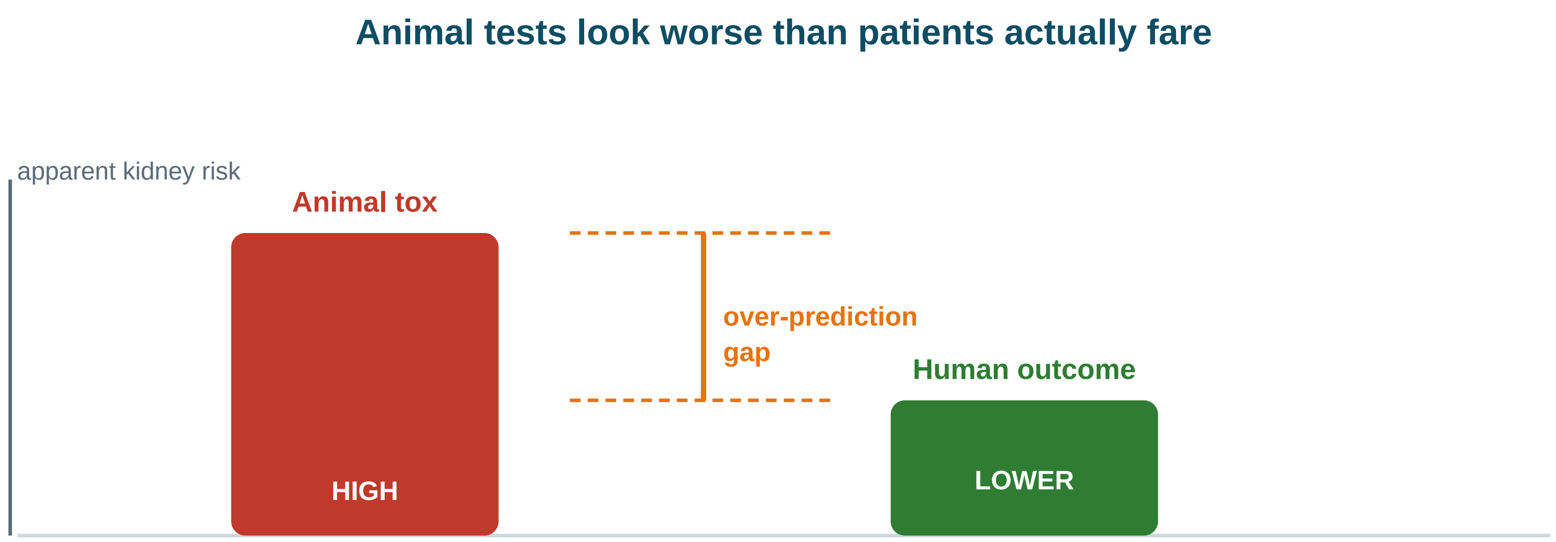
sequences with real letters

13 → 33

tripled — the most model-ready slice

Most sources give a sequence *or* a toxicity value. This public-domain patent table gave **both in the same row** for many oligos — the most directly model-ready slice of the dataset, and it tripled our sequence coverage.

Finding 3 — animal tests over-predict human risk



Human & animal rows sit side by side, measured the same way — so the gap can be **modelled, not hidden**.

A known, consistent bias for this drug class: animal kidneys look worse than patients turn out to be. We captured human and animal rows side by side so the gap can be **modelled, not ignored** — don't treat animal histopathology as human ground truth.

PART 5

Trust, limits, and the ask

Every row is defensible

PROVENANCE — THE PAPER TRAIL

Each measurement carries its **source**, the **exact** table/figure/claim, and its **redistribution right**, so any value can be re-checked at its origin.

47 public-domain rows

64 summary-stat rows

16 sources

QUALITY CONTROL — ALL PASSING

- ✓ every category value from an allowed list
- ✓ column counts intact (17 / 23)
- ✓ every result links to a real drug — **0 orphans**
- ✓ no duplicate IDs · grades all 0–3
- ✓ no-guessing sequence rule held

Honest limitations

- **Grades are provisional** — assigned by rubric, awaiting expert sign-off.
- **Sequences 33/65** — remaining gaps (siRNA guide strands, some PMOs) left blank, never guessed.
- **36 “WS” rows** rest on web summaries — need verification against the primary document.
- **In-vitro human-cell rows (19)** are the scientific core but still a minority — growing them is the top priority.
- **Animal over-prediction** is present by design — it must be modelled, not ignored.

Stating the soft spots plainly is part of the deliverable — a reviewer should know exactly where to push.

What we need from German

1. **Grade sign-off.** Check the rubric → row mapping in `measurements.csv` so we can remove the “provisional” flag. Hot spots: the **grade 2 ↔ 3** line (biomarker rise vs. true AKI) and the **patent words → grade** mapping.

2. **Biology sanity check.** Is the functional-vs-structural story (megalin/cubilin → protein leak → grade 1) faithful, and are the readout → severity calls physiologically sound?

3. **Source confidence.** Flag any anchor — especially the 36 “WS” rows — you’d want re-verified against the primary text before release.

On hold until sign-off: the ≤12-page narrative · verifying WS rows · backfilling sequences · mining patent N4 for more rows.

Roadmap

DONE

- 111 graded, kidney-specific rows (≥100 target met)
- All 7 oligo families; full 0–3 severity range
- Two-table schema, full provenance, QC passing
- Methodology + this deck

NEXT (POST SIGN-OFF)

- Finalise grades → remove provisional flag
- Verify the 36 WS rows against primary text
- Backfill more sequences; mine patent N4
- Write the ≤12-page narrative for submission

Submission window: **1 May – 31 Dec 2026**. The dataset already clears the volume bar; the remaining work is verification and write-up.

Where everything lives

```
README.md          strategy, scope, live record counter
schema.md          data dictionary + the grade rubric
METHODOLOGY.md     how it was built (sources → grading → QC)
PRESENTATION.md    this deck (+ plain-English speaker notes)
data/oligos.csv    65 drugs · 17 design columns
data/measurements.csv  111 graded rows ← review here
sources/           the source PDFs + the source registry
```

Thank you — feedback welcome at the row level. Every number in this deck regenerates from `data/`; nothing here is hand-maintained prose detached from the tables.